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SEAT No. _____

No. of printed pages : 02

Sardar Patel University
External Examination (CBCS)
B.Sc. (C.A. & IT) / M.Sc. Information Technology (Integrated)
4th Semester (CBCS)
PS04CIIT02 : Data Communication & Networking
23rd March, Saturday - 2019

Time : 02:00 pm to 05:00 pm

Total Marks : 70

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Q.1 Select the appropriate option.

- When the packets are small and all the same size, they are often called _____.
(a) Cells (b) Trunks (c) Circuits (d) Subnet
- If a computer on the network shares resources for others to use, it is called _____.
(a) Server (b) Client (c) Mainframe (d) Router
- Which of the following is not a guided medium?
(a) Twisted-pair cable (b) Coaxial cable
(c) Fiber optic cable (d) Atmosphere
- Which of the following modulation requires the lowest bandwidth?
(a) ASK (b) PSK (c) FSK (d) QPSK
- Which layer is not present in OSI Model?
(a) Internet Layer (b) Transport Layer
(c) Physical Layer (d) Application Layer
- TCP/IP _____ layer corresponds to the OSI models to three layers.
(a) Application (b) Presentation (c) Session (d) Transport
- Which of the following is not a valid form of CSMA?
(a) 1-Persistent (b) P- Persistent (c) Non- Persistent (d) Persistent
- Which protocol ensures reliable delivery?
(a) TCP (b) UDP (c) both a & b (d) none of these
- A Bluetooth network can have _____ master(s).
(a) one (b) two (c) three (d) eight
- IPv6 has _____ bit addresses.
(a) 32 (b) 64 (c) 128 (d) 8

Q.2 Answer the following questions. **(Attempt any TEN)**

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- What are the difference between Router & Switch?
- How can we classify Computer Networks?
- List the transmission modes.
- Explain amplitude modulation in brief with example.
- Explain serial transmission in brief.
- Explain phase modulation with in brief example.

(P.T.O.)

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7. What is TCP? 5
 8. Explain P-Persistent in brief. 5
 9. Draw figure for IPv4 header. 5
 10. What is satellite? List categories of satellite. 5
 11. What is Physical topology and Logical topology? 5
 12. Explain Infrared Transmission in brief. 5
- Q.3** [a] Define computer network. State the advantages disadvantages of computer network. 5
- [b] Write detailed note on Wide Area Network. 5
- OR**
- Q.3** [a] Discuss in detail Fiber Optics Transmission Media and its advantages. 5
- [b] Explain STAR topology and MESH topology with their advantages and disadvantages. 5
- Q.4** [a] What is Switching? Explain Packet Switching in detail. 5
- [b] Explain OSI Reference Model in detail. 5
- OR**
- Q.4** [a] What do you mean by Multiplexing? Explain FDM in detail. 5
- [b] Explain TCP/IP Reference Model. 5
- Q.5** [a] Write short note on CSMA. 5
- [b] Explain TCP Connection Establishment. 5
- OR**
- Q.5** [a] Explain TCP Segment Header. 5
- [b] Write sort note on Classful Addressing Scheme. 5
- Q.6** [a] Write a note on Geosynchronous Satellite. 5
- [b] Write a short note on Microwave Transmission. 5
- OR**
- Q.6** [a] Write a short note on Wireless LAN and Wireless WAN. 5
- [b] Explain Communication Satellites in detail. 5

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